

Operating System

ICT1308

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CHAPTER I-

Introduction

Computer System

- Computer system is a collection of hardware and software components designed to provide an effective tool for computation.
- Hardware refers to electrical, mechanical, and electronic parts that make up the computer.
- Software refers to the set of programs written to provide services to the system. It gives life and meaning to hardware and brings it to the operational level.
- Software's are basically two types
 1. Application software
 2. System Software

Computer System

1. Application software

- Set of programs written for a specific area of application.
- Examples are word processors, spreadsheets, and database management systems, Educational software, multimedia software

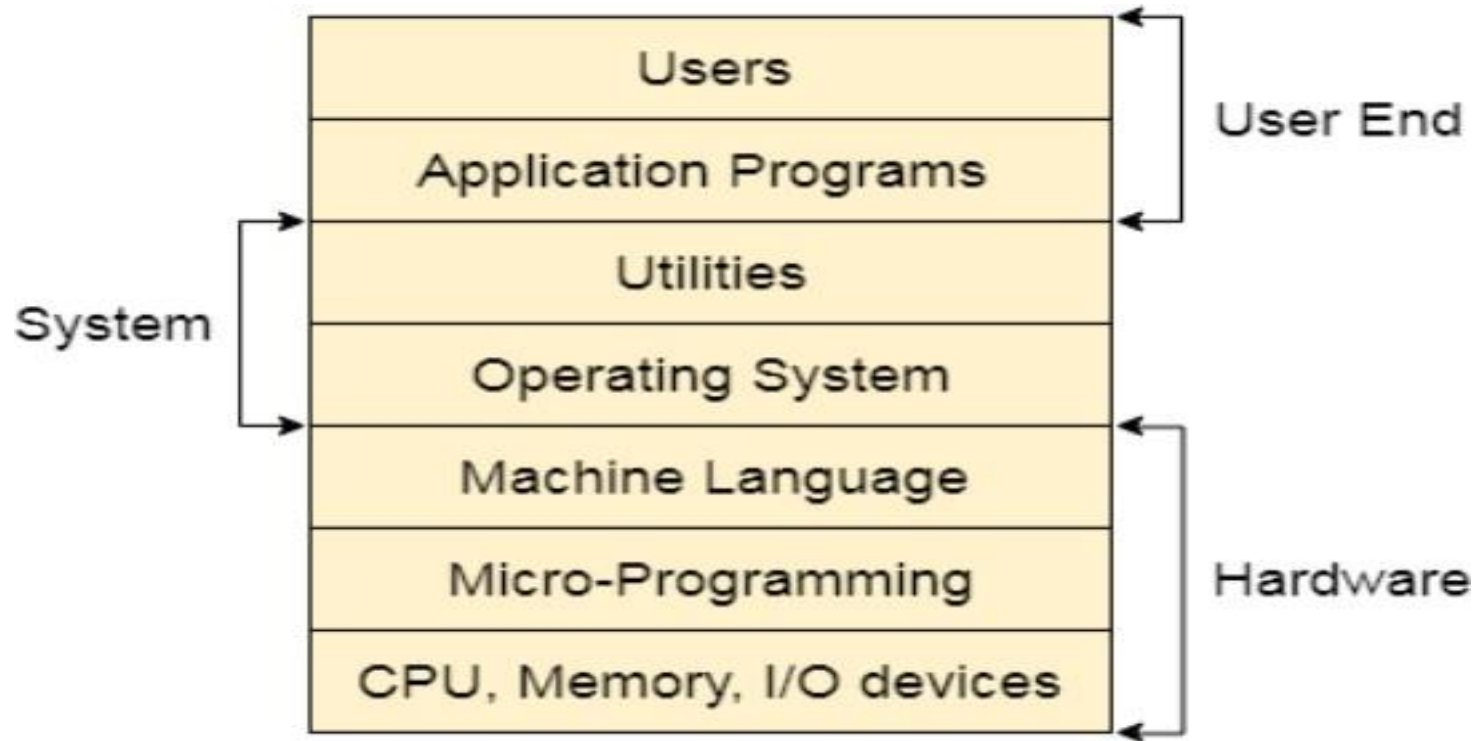
2. System software

- Set of programs written for running computer hardware and applications and managing its resources such as memory, processors, and resources.
- Provide a platform for running application software.
 - Operating systems, device drivers, antivirus programs, utility programs

Structure of a computer system

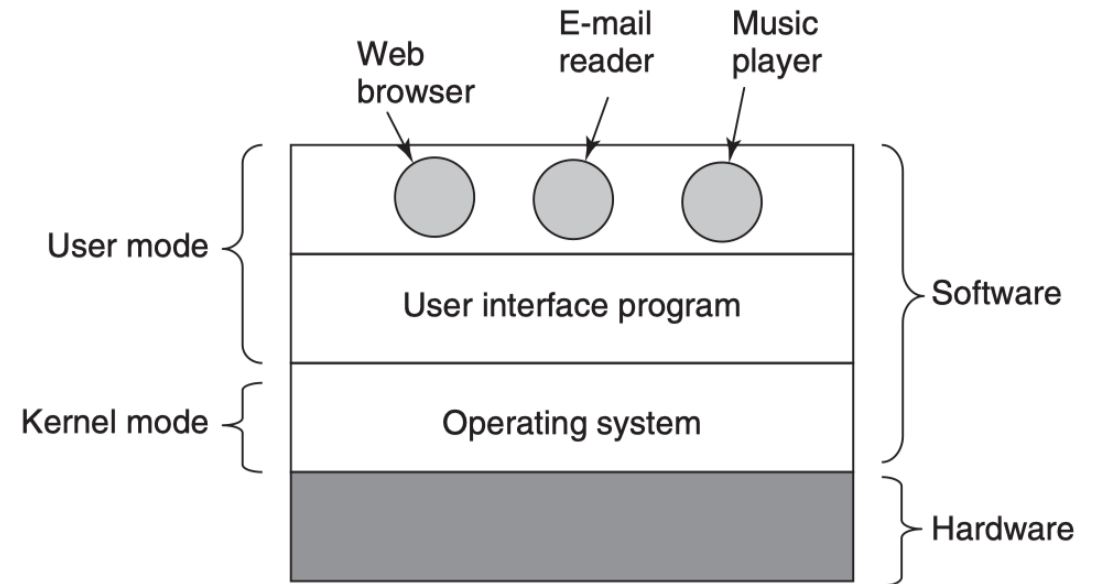
- Computer system consists of:
 - **Users** -People who use the computer
 - **Application programs**-Compilers,Databases, Games, Video players, browsers, etc.
 - **System programs**-Shells, Editors, Compilers, etc.
 - **Operating system** -Special program which acts as a interface between the user and hardware
 - **Hardware**-CPU, Disk, memory, etc.

Structure of a computer system



Where the OS Fits In

- Two modes of operation:
 - **Kernel mode / Supervisor mode**
 - In this mode it has complete access to all h/w and execute any instruction machine is capable of executing.
 - **User mode** – Other SW runs in user mode. Subset of machine instructions are available to user mode. Instructions that affect control of the machine or do I/O are forbidden to user-mode programs.



What is an Operating System?

- OS is a program that acts as an intermediary between a user of a computer and the computer hardware.
- An OS is a collection of system programs that together control the operation of a system.

Examples-UNIX, MS-DOS, MS-Windows, Windows/NT, MacOS

- Operating systems indeed perform two fundamentally distinct yet equally important functions :

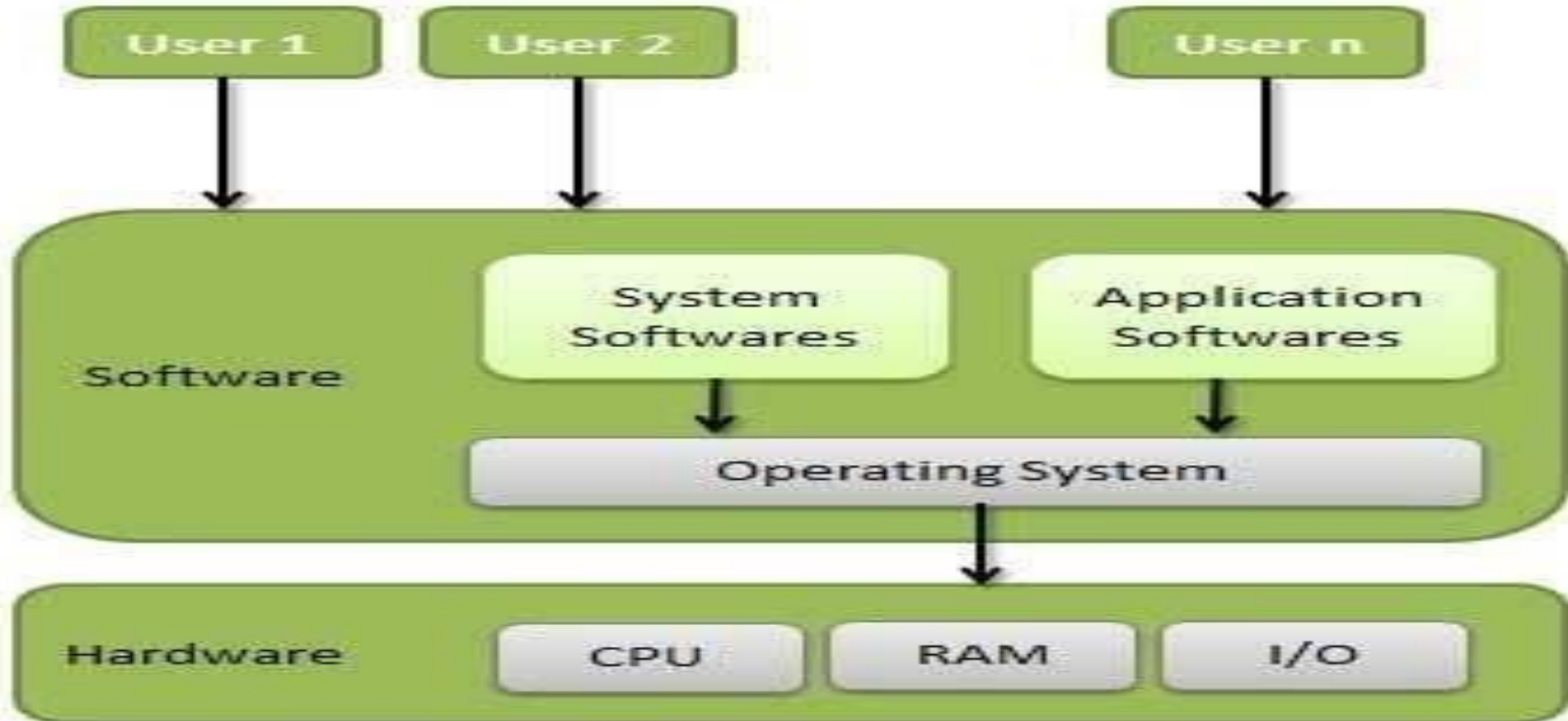
1. The Operating System as an Extended Machine

In this role, the operating system acts as a Virtual machine, **abstracting away the complexities of the underlying hardware** and providing a simplified, easier-to-use interface to both programmers and users. The messy, intricate details of hardware are hidden behind layers of abstraction, making it easier to interact with the system without needing to manage low-level operations. In essence, the OS is like a translator or intermediary that **makes the hardware user-friendly**

2. The Operating System as a Resource Manager

While the OS simplifies the interface for users and applications, it simultaneously manages the **hardware resources(CPU, memory, devices)** of the computer to ensure that they are used efficiently and fairly. This role is crucial for preventing resource conflicts, ensuring that no one program hogs all the system resources, and enabling multiple processes to run concurrently.

Structure of Operating System



History of Operating Systems

1. Prehistory of operating systems (1940 to Early 1950)

- No formal Operating systems
- These early machines were entirely **manual systems**, and users needed to input data using **punch cards** or **plugboards**.
- The programs had to be written specifically for each machines, and there was no concept of software layers or abstraction.
- The **ENIAC** (Electronic Numerical Integrator and Computer) is an example of an early computer without an OS. Operators had to physically connect and set up the machine for each problem.

2. First Generation of Operating Systems: Batch Systems (1950s to Early 1960s)

- Batch processing: Jobs were collected, or “batched,” together and processed sequentially.
- The OS would load a job from a batch, execute it, and move on to the next without manual intervention.
- Early batch systems used a monitor program to manage the job queue.
- IBM 701 and 7094 are examples

3. Second Generation of Operating Systems: Multiprogramming and Time-Sharing (1960s to 1970s)

- During the 1960s, computers became powerful enough to support multiple users simultaneously.
- This led to the development of multiprogramming and time-sharing systems, where multiple programs or users could run on the same machine by sharing system resources.
- Introduction of interrupts and job scheduling algorithms for better resource management.
- CTSS (Compatible Time-Sharing System) and MULTICS (Multiplexed Information and Computing Service)

4. Third Generation General-Purpose Operating Systems (1970s to 1980s)

- The birth of operating systems that supported minicomputers and personal computers
- General-purpose systems: Operating systems were developed to run on different hardware platforms, breaking away from single-purpose systems.
- Support for virtual memory, which allowed programs to use more memory than was physically available by paging parts of programs in and out of physical memory.
- Increased emphasis on portability: OS could be moved to different hardware architectures with minimal modifications.
- UNIX designed as a portable, multi-tasking, multi-user operating system.

5. Fourth Generation: PC Operating Systems and GUIs (1980s to 1990s)

- Personal computers: Operating systems were developed specifically for desktop machines used by individuals.
- Introduction of GUIs: GUIs replaced command-line interfaces (CLIS) to make systems more user-friendly.
- Integration of networking capabilities, leading to the early stages of networked computing and the rise of the internet.
- MS-DOS, Windows 1.0 and Apple macintosh

6. Fifth Generation: Modern Operating Systems (1990s to present)
- Operating systems evolved rapidly to accommodate advancements in hardware, networking, and user demands.
 - The growth of internet, mobile computing, and the rise of distributed systems and virtualization have all shaped modern operating systems
 - Windows NT, Linux, macOS, Android and iOS

7. Sixth Generation: Cloud, Virtualization, and Distributed Systems (2000s to Present)

- Cloud computing: Many applications now run on remote servers (in the cloud), with operating systems managing large, distributed infrastructures. Examples include Amazon Web Services (AWS) and Google Cloud.
- Virtualization: Technologies like VMware, KVM, and Hyper-V allow multiple operating systems to run on the same physical hardware, facilitating cloud services, development, and testing.
- Containerization: Systems like Docker and Kubernetes allow applications to be deployed in isolated containers that share the OS kernel but run independently

Types of Operating Systems

Types of Operating System



Types of Operating Systems

- **Batch Operating System**

- Access is given to more than one person; they submit their respective jobs to the system for execution.
- The system put all of the jobs in a queue on the basis of first come first serve and executes jobs one by one.
- Users get their respective outputs when all the jobs get executed.
- Purpose of this OS was to transfer the control from one job to another as soon as the job was completed.

Types of Operating Systems

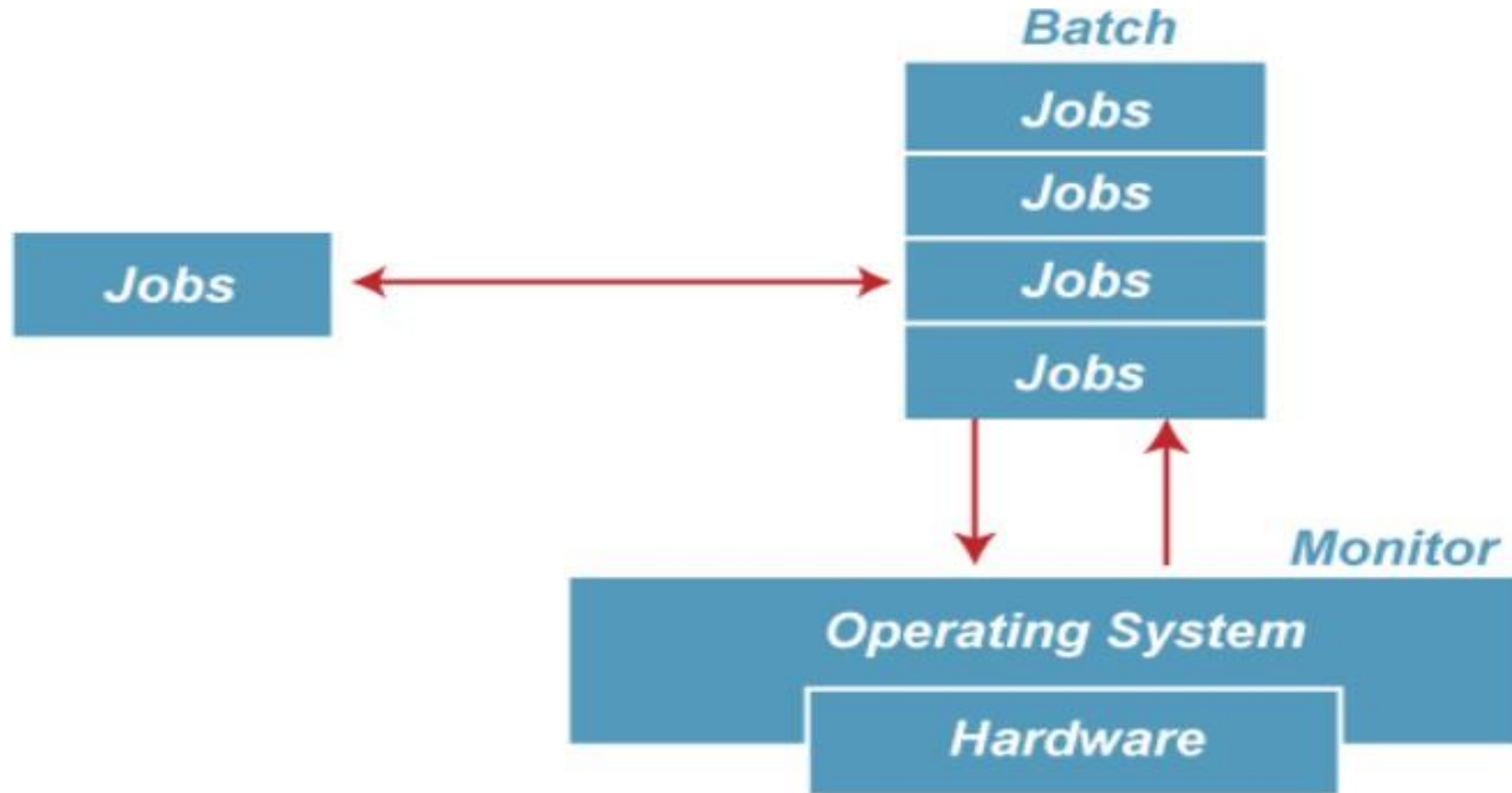
Advantages

- The use of a resident monitor improves computer efficiency as it eliminates CPU time between two jobs.

Disadvantages

- Starvation
- Not interactive

Types of Operating Systems



Types of Operating Systems

- **Multiprogramming OS**

- Extension to batch processing where the CPU is always kept busy.
- Each process needs two types of system time: CPU time and IO time.
- In a multiprogramming environment, when a process does its I/O, the CPU can start the execution of other processes.

Advantages

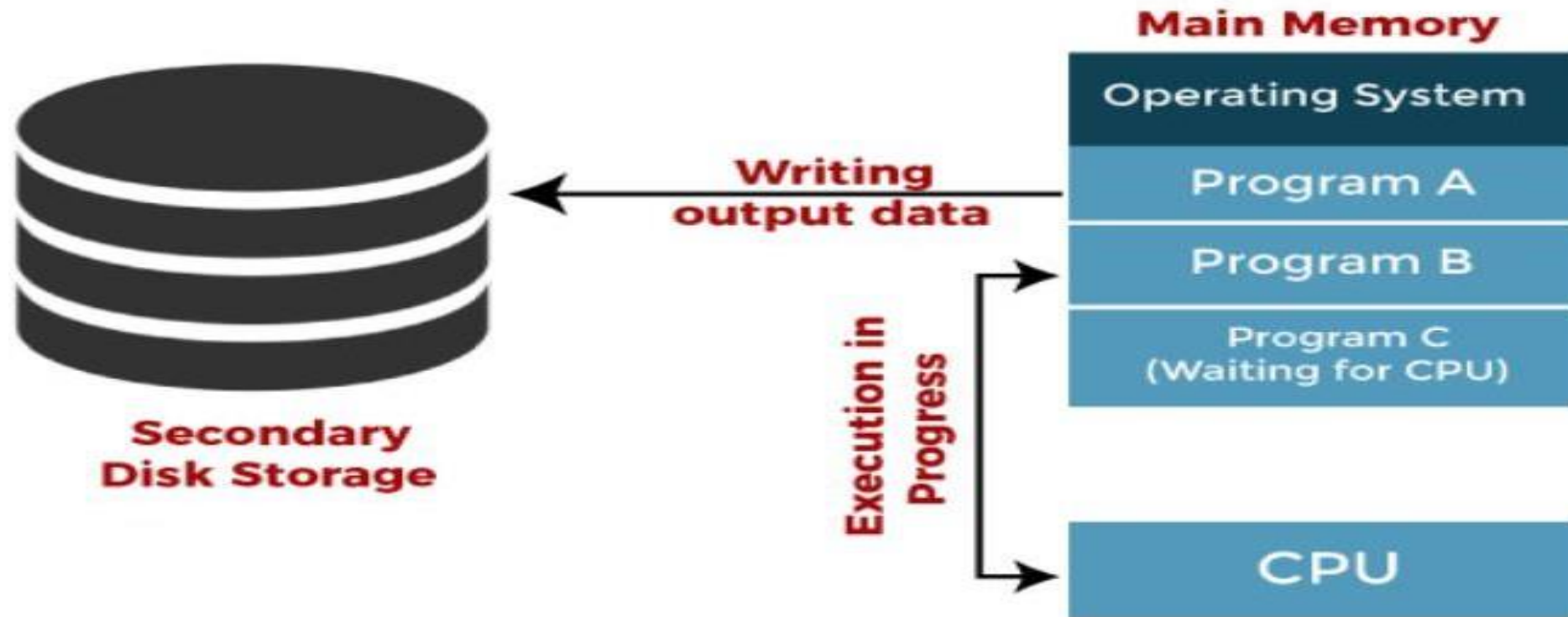
- Throughput of the system is increased as the CPU always had one program to execute.
- Response time can also be reduced.

Types of Operating Systems

Disadvantages

- Various systems resources are used efficiently, but they do not provide any user interaction with the computer system.

Types of Operating Systems



Types of Operating Systems

- **Multiprocessing OS**

- Parallel computing is achieved by multiprocessing.
- There is more than one processor present in the system which can execute more than one process at a time.
- This increases the throughput of the system.

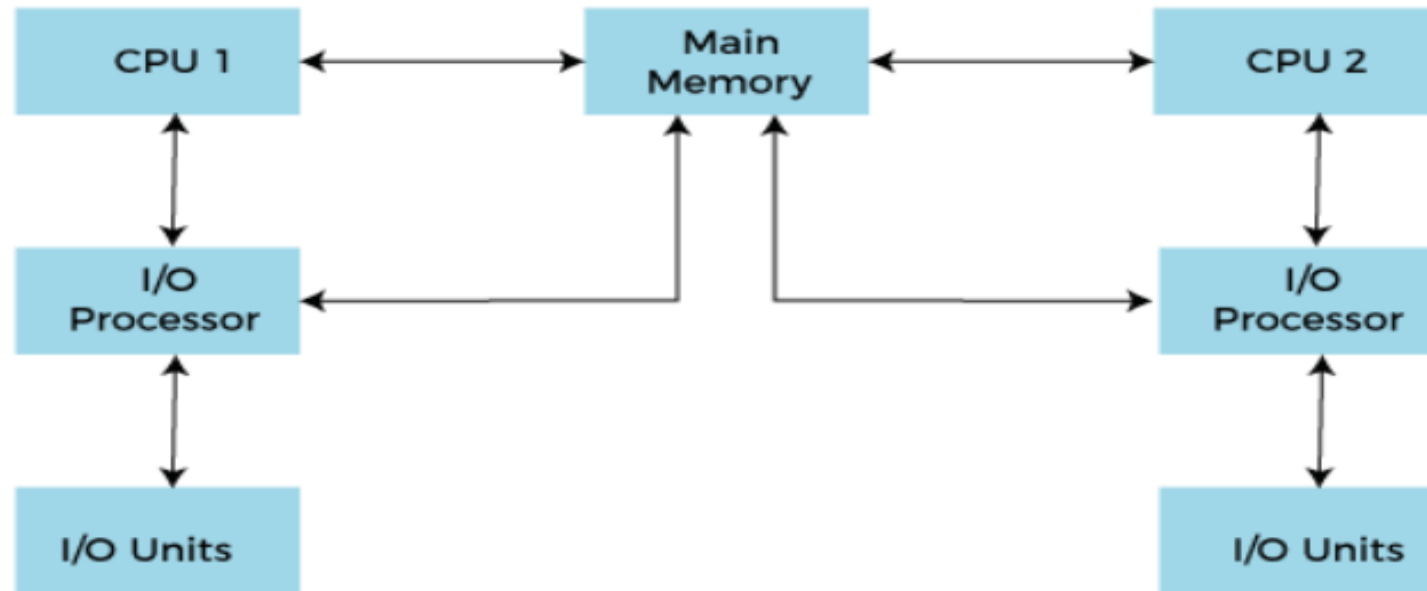
Advantages

- Increased reliability
- Increased throughput

Types of Operating Systems

Disadvantages

- Complex and sophisticated as it takes care of multiple CPUs simultaneously.



Types of Operating Systems

- **Multitasking OS**

- Logical extension of multiprogramming system that enables multiple programs simultaneously.
- It allows a user to perform more than one computer task at the same time.

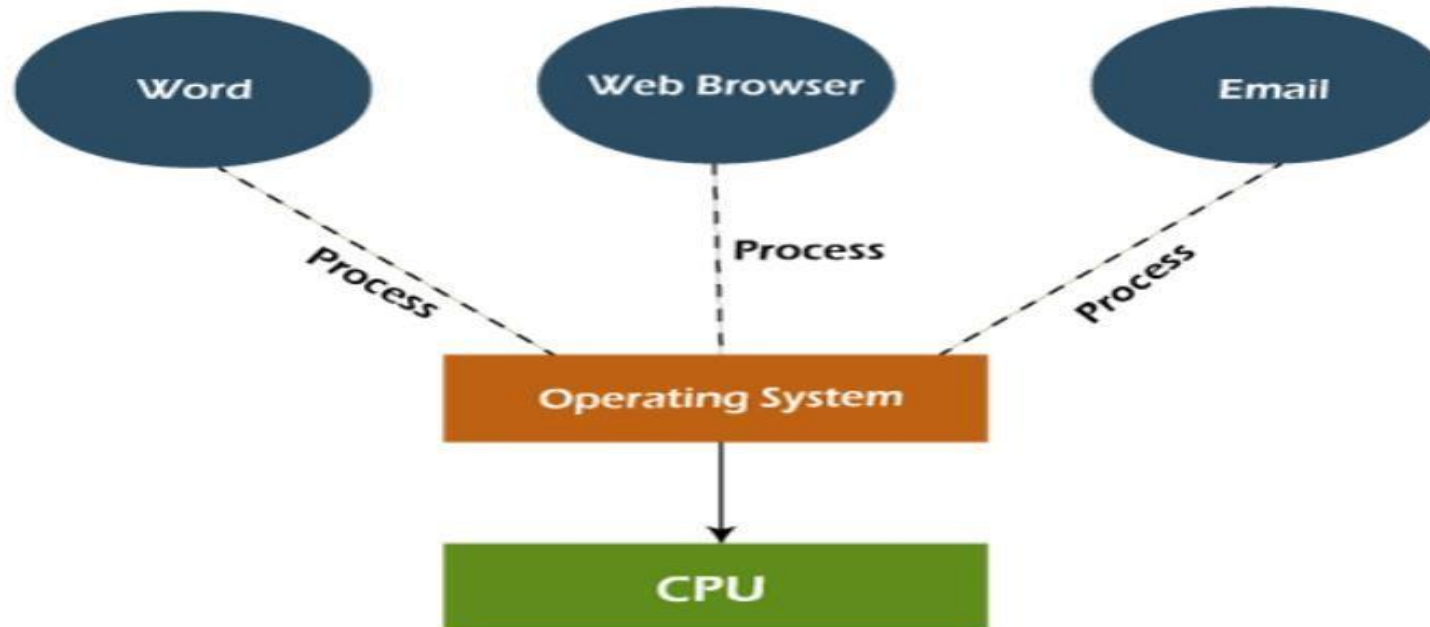
Advantages

- More suited to support multiple users simultaneously.
- Have well defined memory management.

Types of Operating Systems

Disadvantage

- Busier at the same time to complete any task in a multitasking environment, so the CPU generates more heat.



Types of Operating Systems

- **Network Operating System**

- Include software and associated protocols to communicate with other computers via the network conveniently and cost-effectively.

- Types of network OS:

- 1. Peer to peer

- 2. Client-server system

Advantages

- network traffic reduces due to the division between clients and the server.
- This type of system is less expensive to set up and maintain.

Types of Operating Systems

Disadvantages

- Failure of any node in the system affects the whole system.
- Trained network administrators are required.



Types of Operating Systems

- **Real Time Operating System**

-Each job carries a certain deadline within which the job is supposed to be completed, otherwise huge loss will be there or even if the result is produced it will be completely useless.

-The Application of a Real-Time system exists in the case of military applications, if you want to drop a missile, then the missile is supposed to be dropped with a certain precision.

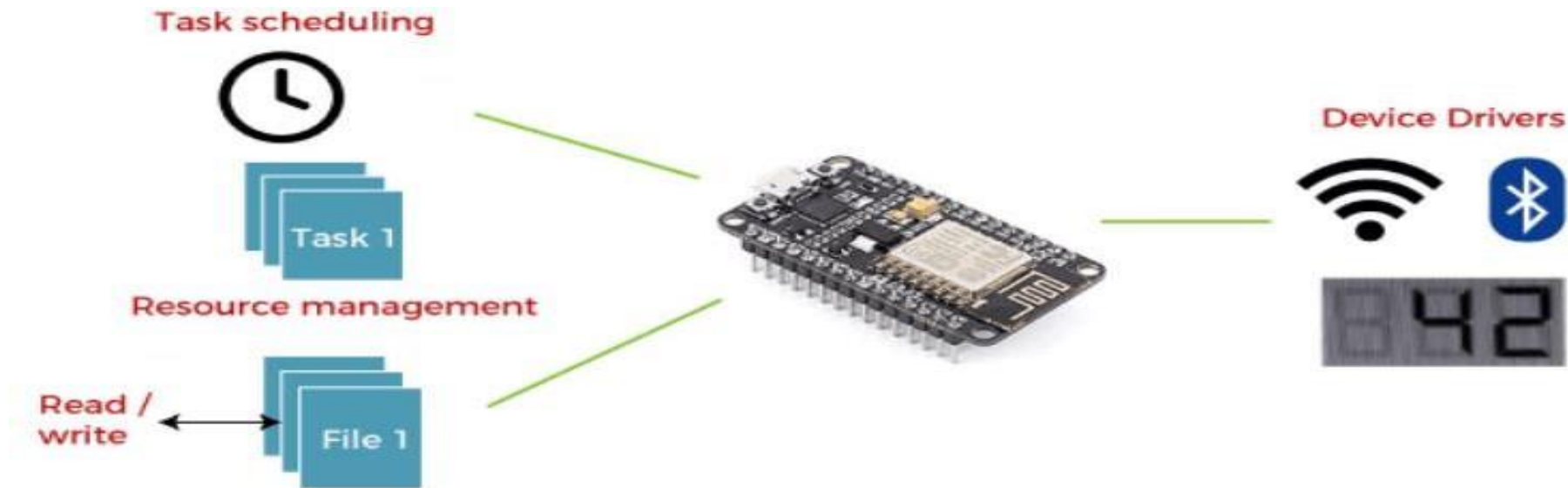
Advantages

- Easy to layout, develop, and execute real-time applications under real-time OS
- Maximum utilization of devices and systems.

Types of Operating Systems

Disadvantages

- Very costly and develop
- Very complex and consume critical CPU cycles.



Types of Operating Systems

- **Time-sharing OS**

- Computer resources are allocated in a time-dependent fashion to several programs simultaneously.

- Thus it helps to provide a large number of user's direct access to the main computer

- It is a logical extension of multiprogramming.

- In time-sharing, the CPU is switched among multiple programs given by different users on a scheduled basis.

Types of Operating Systems

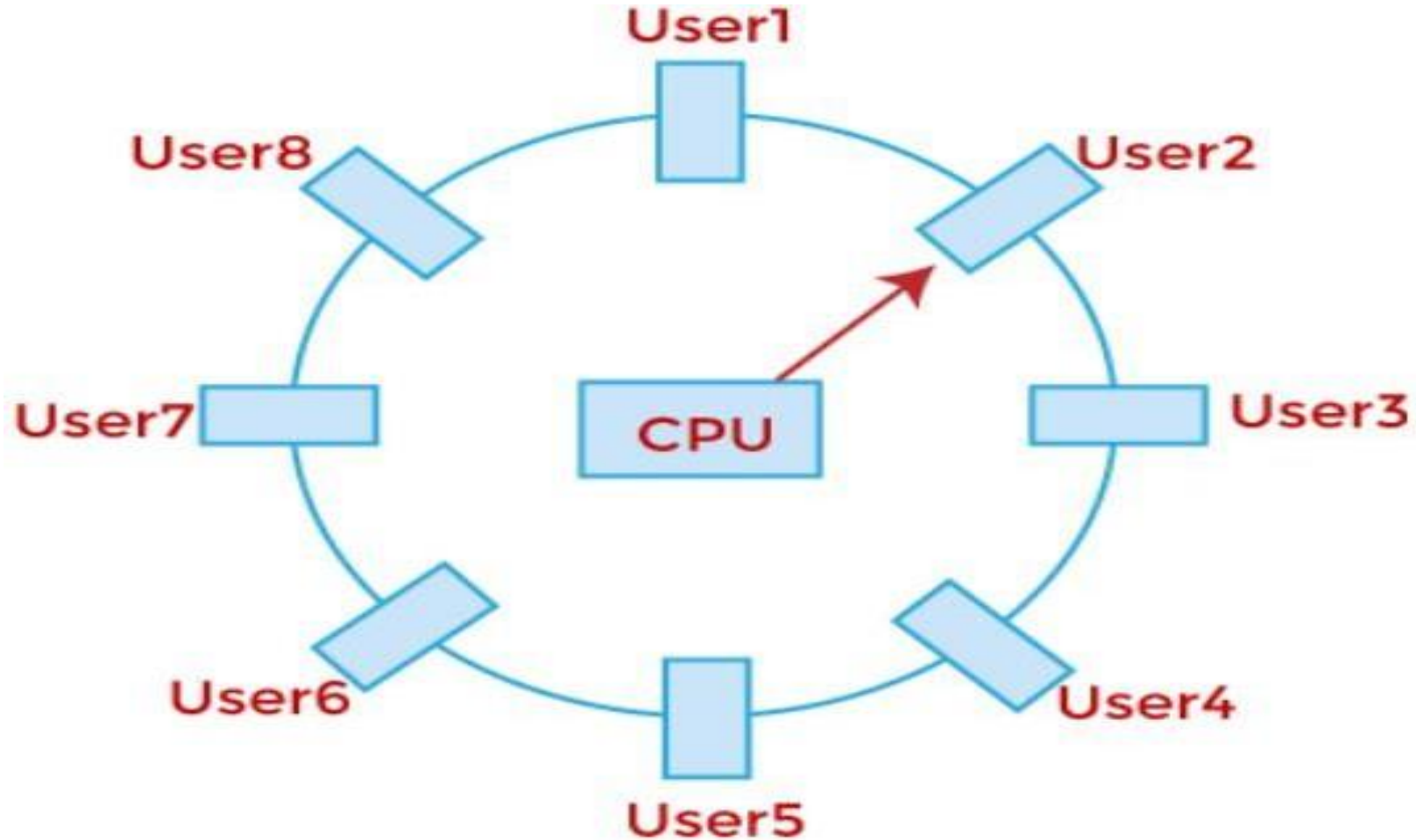
Advantages

- Provide effective utilization and sharing of resources
- System reduces CPU idle and response time

Disadvantages

- Data transmission rates are very high compared to other methods.
- Security and integrity of user programs loaded in memory and data need to be maintained as many users access the system at the same time.

Types of Operating Systems



Types of Operating Systems

- **Distributed OS**

- Not installed in a single machine, distributed into parts and these parts are loaded on different machines.
- A part of the distributed Operating system is installed on each machine to make their communication possible.
- More complex, large, and sophisticated than Network operating systems because they also have to take care of varying networking protocols.

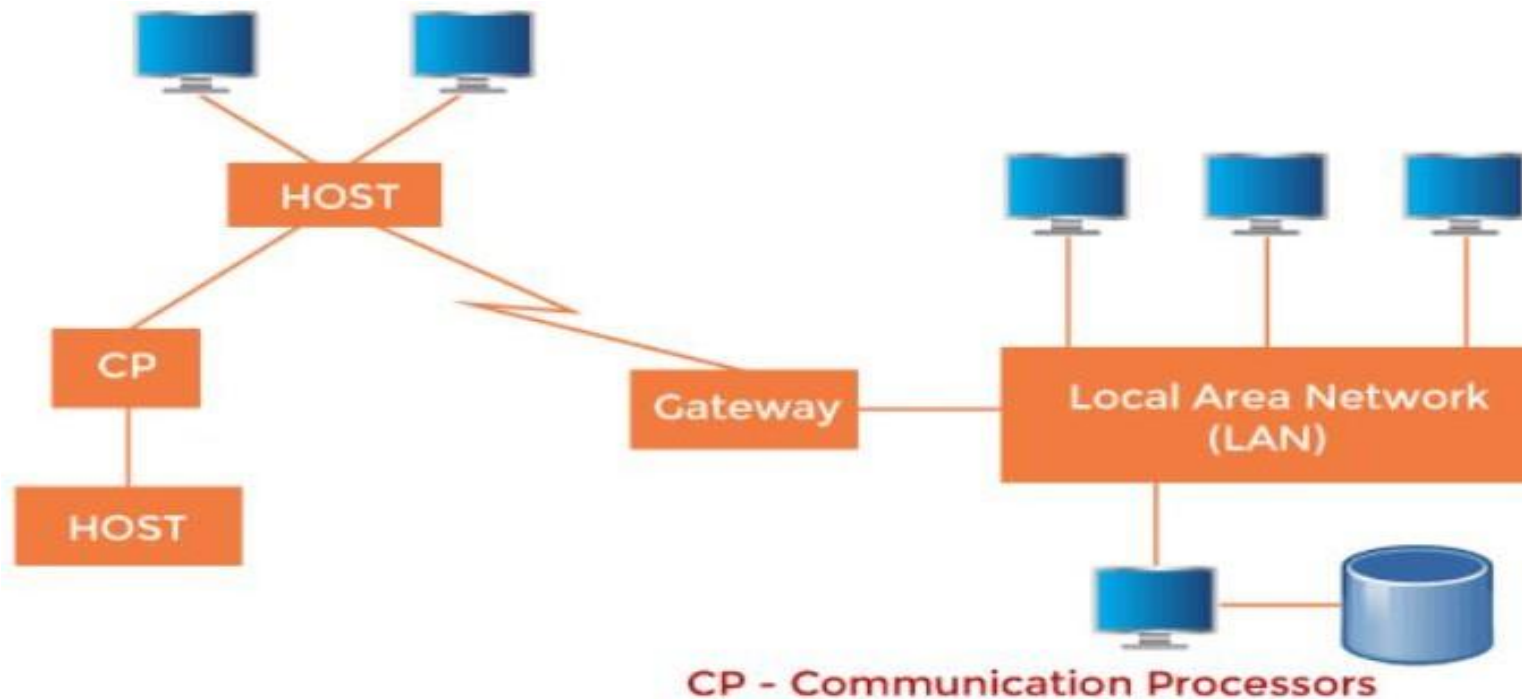
Advantages

- Provide sharing resources
- Fault tolerate

Types of Operating Systems

Disadvantages

- Protocol overhead can dominate computation cost.



Operating System Concepts



Processes

- Program in execution
- Each process has its address space list of memory space from 0 to maximum which the process can read and write.
- The address space contains the executable program, the program's data, and its stack.
- Also associated with each process is a set of resources, commonly including registers (including the program counter and stack pointer), a list of open files, out standing alarms, list of related processes, and all the other information needed to run the program.
- Example – Running a text editor is process, and if you open two instances of the text editor each instance is a separate process.



Address Space

- Every computer has some main memory that it used to hold executing programs.
- In a very simple operating system, only one program at a time is in memory. To run a second program, the first one has to be removed and the second one placed in memory.
- More sophisticated operating systems allow multiple programs to be in memory at the same time.
- Each process has its own address space, which isolates it from other processes. This prevents one process from accidentally or maliciously accessing another process's memory.



Input / Output

- All computers have physical devices for acquiring input and producing output.
- Consequently, every operating system has an I/O subsystem for managing its I/O devices.
- Some of the I/O software is device independent, that is, applies to many or all I/O devices equally well. Other parts of it, such as device drivers, are specific to particular I/O devices.



The shell

- The shell is an interface that allows users to interact with the operating system, typically via command-line input. The shell acts as an intermediary between the user and the OS's kernel.
- The shell interprets user commands (such as creating files, listing directory contents, or launching programs) and passes them to the kernel for execution.
- Types of Shells:
 - Command-Line Interface (CLI): Text-based interface where users type commands.
Examples: Bash (Linux), Command Prompt (Windows), Zsh (macOS).
 - Graphical User Interface (GUI): A more user-friendly, graphical interface where users interact with the system through icons and windows.
Examples: Windows Explorer, macOS.